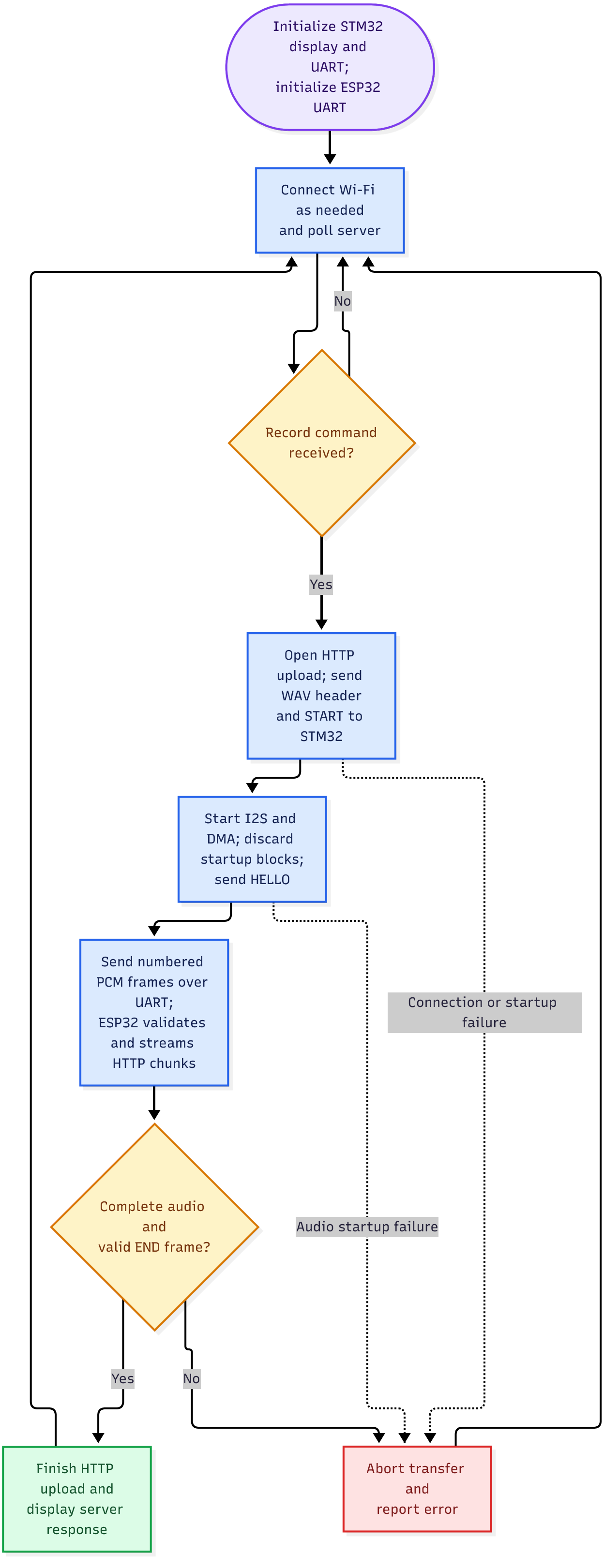

Initialize STM32 display and UART; initialize ESP32 UART
Connect Wi-Fi as needed and poll server
No
Record command received?
Yes
Open HTTP upload; send WAV header and START to STM32
Start I2S and DMA; discard startup blocks; send HELLO
Send numbered PCM frames over UART; ESP32 validates and streams HTTP chunks
Connection or startup failure
Complete audio and valid END frame?
Audio startup failure
Yes
No
Finish HTTP upload and display server response
Abort transfer and report error